

Feasibility-Preserving Multi-Objective Harris Hawks Optimization for Employment-Based Visa Allocation

Javier Augusto Rebull Saucedo, Yazmín Ivonne Flores Martínez, and Raúl Gibrán Porras Alaniz

Universidad Autónoma de Ciudad Juárez,
Maestría en Inteligencia Artificial y Analítica de Datos,
Ciudad Juárez, Chihuahua, México
rebull@outlook.com

Abstract. Employment-based immigration to the United States allocates roughly 140,000 visas per fiscal year under a per-country ceiling and a categorical spillover cascade that together produce decade-long waits, large inter-country disparities, and thousands of unused visas. We cast annual allocation as a three-objective combinatorial program—minimizing unserved waiting load, inter-country disparity, and visa waste—under six hard policy constraints, and solve it with multi-objective Harris Hawks Optimization (MOHHO). The methodological core is a feasibility-preserving decoder: a smallest-position-value random-key mapping composed with a deterministic greedy assignment that satisfies all six constraints by construction, so the swarm never leaves the feasible region and needs no penalty terms or repair. We tune the population size and iteration budget with a Taguchi L9 orthogonal array and a confirmation run. Across 30 runs the recovered Pareto front strictly dominates the incumbent first-in, first-out policy on all three objectives. A controlled six-method ladder then isolates *what* drives performance: the decoder alone (random restart) already beats a real-coded NSGA-II, and—our central finding—the representation-operator match, not the metaheuristic family, governs the result. Matching operators to the representation lifts three distinct paradigms (genetic algorithm, decomposition, and swarm) above every random-key method; the Discrete-MOHHO we introduce is the most stable optimizer. The findings hold across five perturbed-demand instances and replicate on a structurally distinct multidimensional knapsack. The method delivers an auditable menu of allocation policies rather than a single recommendation.

Keywords: Multi-objective optimization · Harris Hawks Optimization · Constraint handling · Random-key decoder · Permutation encoding · Taguchi design of experiments · Resource allocation

1 Introduction

Each fiscal year the United States Congress authorizes approximately 140,000 employment-based (EB) immigrant visas, distributed across five categories (EB-1

to EB-5). Two structural rules govern the distribution: a *per-country ceiling* of 7% of the annual total ($P_c \approx 25,620$ visas) that no single country may exceed, and a *spillover* mechanism that cascades unused visas across categories (EB-4/5 $\rightarrow$ EB-1 $\rightarrow$ EB-2 $\rightarrow$ EB-3). The incumbent system processes petitions strictly first-in, first-out (FIFO) by priority date. The interaction of these rules makes the incumbent policy dysfunctional: two countries concentrate more than 60% of demand and accumulate waits of 10–13 years, while the clash between per-country and per-category ceilings leaves thousands of visas unused.

This matters beyond accounting. Behind every backlogged petition is a household whose relocation, employment, and family reunification hinge on a priority date, so the allocation rule has direct welfare consequences. The problem admits three legitimate, mutually conflicting goals: serve the longest-waiting petitions (*efficiency*), distribute opportunity evenly across nationalities (*equity*), and waste no visas (*utilization*). No single allocation optimizes all three; the policy-relevant object is the set of best compromises—a Pareto front.

Computing that front is hard for three reasons. First, the allocation is *combinatorial*: it assigns integer visa counts to 105 country $\times$ category groups under six simultaneous constraints, and its feasible set generalizes the multidimensional knapsack problem, which is NP-hard [11]. Second, the objectives are genuinely *conflicting*, so the solution is a front rather than a point. Third—and most often underestimated—the *hard constraints* are unforgiving: a continuous population metaheuristic naturally proposes infeasible candidates, and the usual remedies (penalty functions, aggressive repair) either require delicate tuning or distort the very search landscape the metaheuristic explores.

We address these challenges with a multi-objective adaptation of Harris Hawks Optimization (MOHHO) whose centerpiece is a *feasibility-preserving decoder*. Concretely, this paper makes five contributions: (i) a decoder that couples a smallest-position-value (SPV) random-key mapping with a deterministic greedy assignment, guaranteeing all six policy constraints *by construction*—without penalties or repair, and therefore without perturbing the search landscape; (ii) a multi-objective HHO that pairs this decoder with a crowding-pruned external archive and applies it to a real, constraint-heavy allocation problem rather than to synthetic benchmarks; (iii) a Taguchi L9 orthogonal-array tuning of the population size and iteration budget, replacing empirical parameter choices with a systematic experimental design validated by a confirmation run; (iv) a controlled six-method ladder that, by lifting three distinct paradigms (GA, decomposition, and a representation-matched *Discrete-MOHHO* we introduce) to within 0.4% of one another once operators match the representation, shows that the feasibility-preserving decoder and the representation-operator match, not the choice of metaheuristic, govern performance—with a direct operator-order-preservation measurement that explains *why* (interpolating GA operators are near-identity on the decoded permutation); and (v) a rigorous empirical protocol (30 runs, omnibus Kruskal–Wallis and Friedman/Nemenyi tests with effect sizes, strict domination of FIFO, robustness across five perturbed-demand instances, and

a structurally distinct second problem on which the core result replicates) that makes every claim reproducible.

2 Background and Related Work

Harris Hawks Optimization (HHO) [12] is a population-based metaheuristic whose prey *escape energy* E drives a continuous exploration–exploitation transition, with optional Lévy-flight dives. It has been extended to multiple objectives via external archives, elitist non-dominated sorting, and grid indices [14,3,13,5]; these variants are validated mostly on continuous benchmarks, leaving their use on *constrained combinatorial* allocation—where feasibility, not convergence, is the binding difficulty—comparatively unexplored.

We bridge continuous search and combinatorial structure with random keys [2]: a real vector is decoded to a feasible permutation by sorting (the smallest-position-value, SPV, rule). This idea underlies biased random-key genetic algorithms [15] and a general random-key optimizer that pairs a decoder with *any* metaheuristic [4]—the paradigm we instantiate for HHO and probe by varying the metaheuristic while holding the decoder fixed. Among constraint-handling families [17], penalty methods need problem-specific weights and blur feasibility, repair can collapse diversity, and decoder (feasibility-preserving) approaches guarantee feasibility by construction; we adopt the third. NSGA-II [8] is our benchmark and the standard tool for hard allocation and scheduling [22]; we tune parameters with a Taguchi orthogonal array [16,1] and assess fronts with the hypervolume [23], inverted generational distance and spacing [6], nonparametric tests [9], and the A_{12} effect size [20].

3 Problem Formulation

Decision variables. The problem is discretized into $G = C \times |\mathcal{J}| = 21 \times 5 = 105$ groups, where each group g is a (country c , category j) pair. The decision vector is $\mathbf{x} = (x_1, \dots, x_{105})$, $x_g \in \mathbb{Z}_0^+$, where x_g is the number of visas assigned to group g . Each group has a demand (backlog) n_g and a *waiting weight* $w_g = t_{\text{now}} - d_g$, the years elapsed since its priority date d_g .

Objectives. We minimize three conflicting objectives:

$$f_1(\mathbf{x}) = \frac{\sum_g (n_g - x_g) w_g}{\sum_g n_g} \quad \text{unserved waiting load (efficiency),} \quad (1)$$

$$f_2(\mathbf{x}) = \max_{c_1, c_2 \in \mathcal{C}} |\bar{W}_{c_1} - \bar{W}_{c_2}| \quad \text{maximum inter-country disparity (equity),} \quad (2)$$

$$f_3(\mathbf{x}) = V - \sum_g x_g \quad \text{wasted (unassigned) visas (utilization),} \quad (3)$$

where V is the total available visas and $\bar{W}_c = (\sum_{g: c(g)=c} x_g w_g) / (\sum_{g: c(g)=c} x_g)$ is the visa-weighted mean wait of country c . Thus f_2 measures the gap between the longest- and shortest-waiting nationalities: the smaller f_2 , the more equitable the distribution.

Constraints. The feasible set $\mathcal{F}$ is defined by six hard constraints:

- R1: $\sum_g x_g \leq V = 140,000$ (total visas)
- R2: $\sum_{g: c(g)=c} x_g \leq P_c = 25,620 \quad \forall c$ (7% per-country ceiling)
- R3: $\sum_{g: j(g)=j} x_g \leq K_j^{\text{eff}} \quad \forall j$ (effective per-category cap)
- R4: $0 \leq x_g \leq n_g \quad \forall g$ (do not exceed demand)
- R5: $x_g \in \mathbb{Z}_0^+ \quad \forall g$ (integrality)
- R6: FIFO order within each (c, j) queue (seniority within a group)

where K_j^{eff} is the per-category cap after the spillover cascade. The result is a three-objective integer program with six constraints whose feasible space grows combinatorially with G , motivating a metaheuristic rather than exact enumeration.

Pareto optimality. A solution $\mathbf{x}$ *dominates* $\mathbf{x}'$ ($\mathbf{x} \prec \mathbf{x}'$) iff $f_i(\mathbf{x}) \leq f_i(\mathbf{x}')$ for all i and $f_i(\mathbf{x}) < f_i(\mathbf{x}')$ for some i . The *Pareto front* is $\mathcal{P}^* = \{\mathbf{x} \in \mathcal{F} \mid \nexists \mathbf{x}' \in \mathcal{F} : \mathbf{x}' \prec \mathbf{x}\}$. MOHHO approximates $\mathcal{P}^*$ through an external archive.

4 Methods

4.1 Multi-objective Harris Hawks Optimization

Each hawk is a position $\mathbf{H} \in \mathbb{R}^{105}$ in the box $[0, 1]^{105}$. At iteration t the escape energy is

$$E = 2E_0\left(1 - \frac{t}{T}\right), \quad E_0 \sim U(-1, 1), \quad (4)$$

which drives the choice among six movement operators selected by $|E|$ and uniform draws $q, r \in U(0, 1)$ (Table 1). The multi-objective adaptation introduces three elements. First, the *leader* (“rabbit”) is not a single global best but a solution drawn from the external archive with probability proportional to its crowding distance, which biases search toward sparsely populated regions of the front. Second, an *external archive* stores non-dominated solutions and, when it overflows, is pruned by removing the solution with the smallest crowding distance. Third, the archive is updated after every move using the non-dominance relation. The archive size is treated as a tunable parameter (Section 5).

4.2 A feasibility-preserving decoder

The HHO operators act in the continuous space $\mathbb{R}^{105}$, but the problem is combinatorial with six hard constraints. We bridge the two with the SPV rule [2] followed by a deterministic greedy assignment (Algorithm 1). The SPV rule turns a hawk $\mathbf{H}$ into a permutation $\boldsymbol{\pi}$ by sorting the group indices in ascending order of their key h_g . The greedy decoder then walks the groups in that order and assigns

$$x_g = \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}}), \quad (5)$$

Table 1. The six MOHHO movement operators, selected by the escape energy $|E|$ and uniform draws q, r . The archive’s leader supplies the prey position.

Op.	Condition	Behavior
OP1	$ E \geq 1, q \geq 0.5$	Exploration from a random hawk in the swarm.
OP2	$ E \geq 1, q < 0.5$	Exploration relative to the population centroid.
OP3	$0.5 \leq E < 1, r \geq 0.5$	Soft besiege: gradual approach to the prey.
OP4	$ E < 0.5, r \geq 0.5$	Hard besiege: aggressive contraction.
OP5	$0.5 \leq E < 1, r < 0.5$	Soft besiege with rapid Lévy dives.
OP6	$ E < 0.5, r < 0.5$	Hard besiege with rapid Lévy dives.

Algorithm 1 Feasibility-preserving SPV + greedy decoder

Require: hawk $\mathbf{H} \in \mathbb{R}^{105}$; demands n_g ; budgets $V, \{P_c\}, \{K_j^{\text{eff}}\}$

- 1: $\pi \leftarrow \text{ARGSORTASCENDING}(\mathbf{H})$ ▷ SPV: keys $\rightarrow$ permutation
- 2: $V_{\text{rem}} \leftarrow V; P_c^{\text{rem}} \leftarrow P_c \forall c; K_j^{\text{rem}} \leftarrow K_j^{\text{eff}} \forall j$
- 3: **for** $g \in \pi$ **do** ▷ visit groups in SPV order
- 4: $x_g \leftarrow \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}})$
- 5: $V_{\text{rem}} \leftarrow V_{\text{rem}} - x_g; P_{c(g)}^{\text{rem}} \leftarrow P_{c(g)}^{\text{rem}} - x_g; K_{j(g)}^{\text{rem}} \leftarrow K_{j(g)}^{\text{rem}} - x_g$
- 6: **end for**
- 7: **return** $\mathbf{x}$ ▷ satisfies R1–R6 without repair

decrementing the residual budgets of R1–R3 after each step. Because every assignment is capped by the remaining budgets and by demand, and counts are integers, *every* decoded vector satisfies R1–R6 by construction. No penalty term is introduced and no repair is performed, so the objective landscape the hawks explore is exactly the feasible-region landscape—preserving both the structure of HHO’s cooperative foraging and its stochasticity.

Proposition 1 (Feasibility by construction). *For any hawk $\mathbf{H} \in \mathbb{R}^{105}$, the allocation $\mathbf{x}$ returned by Algorithm 1 satisfies all six constraints R1–R6.*

Proof. The residuals are initialized to $V, P_c, K_j^{\text{eff}} \geq 0$ and, at each step, decremented by $x_g = \min(n_g, V_{\text{rem}}, P_{c(g)}^{\text{rem}}, K_{j(g)}^{\text{rem}})$. By induction each residual stays ≥ 0 , since x_g never exceeds the current residual; hence $0 \leq x_g \leq n_g$ (R4) and, as the minimum of nonnegative integers, $x_g \in \mathbb{Z}_0^+$ (R5). Because $x_g \leq V_{\text{rem}}$ and V_{rem} is the unconsumed total, $\sum_g x_g \leq V$ (R1); likewise $x_g \leq P_{c(g)}^{\text{rem}}$ and $x_g \leq K_{j(g)}^{\text{rem}}$ keep the per-country and per-category sums within P_c (R2) and K_j^{eff} (R3). Finally, each group is a single (c, j) pair sharing one priority date, so assigning the count x_g serves its x_g most-senior petitions, honoring FIFO order within the queue (R6). Thus $\mathbf{x} \in \mathcal{F}$ for every input, with no penalty term and no repair. $\square$

4.3 Benchmarks and a representation-matched optimizer

To separate the contribution of the decoder, the representation, and the search heuristic, we compare a *ladder* of methods that all share the same greedy decoder,

objectives, hypervolume reference point, and 50×500 budget; they differ only in how they explore. Two operate on the continuous SPV random keys in $\mathbb{R}^{105}$: (a) the MOHHO of Section 4.1; and (b) **NSGA-II** [8] with simulated binary crossover (SBX, $\eta_c = 20$, $p_c = 0.9$), polynomial mutation ($\eta_m = 20$, $p_m = 1/d$), binary tournament selection, and elitist ($\mu + \lambda$) replacement. A third, (c) **random restart**, draws the 25,000 candidates uniformly at random (no search), isolating what the decoder alone achieves.

The ablation (Section 6.3) reveals that the continuous random-key operators are a poor match for the induced permutation landscape. We therefore also evaluate three **permutation-native** methods—spanning three distinct paradigms—that act directly on permutations of the 105 groups (decoded by the same greedy procedure): (d) **permutation-NSGA-II** (dominance-based) with order crossover (OX) and swap mutation; (e) **permutation-MOEA/D** [21] (decomposition-based), with Tchebycheff scalarization over a simplex-lattice of weight vectors and the same OX + swap operators; and (f) **Discrete-MOHHO**, our representation-matched Harris Hawks optimizer. Discrete-MOHHO keeps HHO’s signature—the escape energy E of Eq. (4) scheduling exploration and exploitation, and a crowding-pruned external archive supplying the leader—but expresses every move on permutations: when $|E| \geq 1$ a hawk explores by recombining (OX) with a random hawk or by a heavy random shuffle; when $|E| < 1$ it besieges the leader by inheriting an order segment from it (OX) and perturbing with a number of swaps that shrinks as $|E| \rightarrow 0$, with occasional Lévy-like segment reversals. All six methods run for 30 independent seeds under the identical budget.

4.4 Evaluation protocol and indicators

We assess each approximation front with four indicators: the *hypervolume* (HV; larger is better) dominated with respect to the reference point (10; 16; 50,000), an upper bound on each objective; the *inverted generational distance* (IGD; smaller is better) to a reference front Z formed by the non-dominated union of the MOHHO and NSGA-II combined fronts ($|Z| = 444$); the *spacing* indicator (smaller is better) measuring uniformity; and the *cardinality* of the non-dominated set. Objectives are min–max normalized before IGD and spacing. Each algorithm is run for 30 independent seeds, and a *combined front* is formed as the non-dominated union of its 30 per-run fronts. Statistical significance between the two 30-value HV distributions is assessed with a one-sided Mann–Whitney U test [9], and effect size with the Vargha–Delaney A_{12} [20]. The deterministic **FIFO** baseline orders groups by ascending priority date and is decoded with the same greedy procedure, yielding $f_1 = 7.2138$, $f_2 = 12.6377$ years, and $f_3 = 17,540$ wasted visas.

5 Parameter Tuning by Taguchi Orthogonal Array

We replace empirical parameter choices by a Taguchi $L9(3^4)$ orthogonal-array design [16,1] over four factors at three levels—population $N \in \{30, 50, 70\}$,

Table 2. Response table: mean S/N (dB) per factor level, range Δ , and rank. Larger Δ means a more influential factor; the highest-S/N level in each row is in bold (levels tied to rounding are both bolded).

Factor	Level 1	Level 2	Level 3	Δ	Rank
A: population N	124.879	125.034	125.086	0.207	1
B: iteration budget T	124.896	125.042	125.061	0.166	2
C: archive size	125.014	124.971	125.014	0.043	4
D: Lévy exponent β	125.027	125.011	124.961	0.067	3

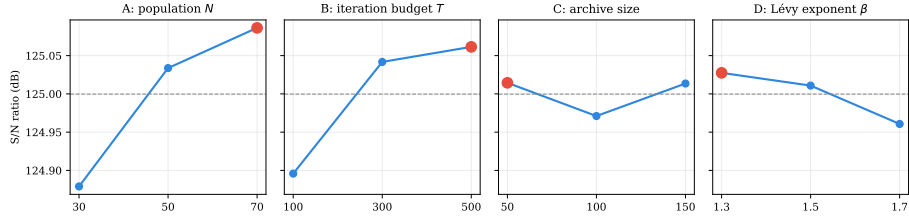


Fig. 1. Main-effects plot of the S/N ratio. The dashed line is the grand mean (125.00 dB) and the red marker the best level per factor; population and iteration budget show clear, saturating trends, while archive size and β are essentially flat.

iteration budget $T \in \{100, 300, 500\}$, archive size $\in \{50, 100, 150\}$, and Lévy exponent $\beta \in \{1.3, 1.5, 1.7\}$ —with the larger-the-better signal-to-noise (S/N) ratio of the hypervolume as response, each of the nine configurations replicated over 12 seeds disjoint from the evaluation seeds.

Averaging the S/N ratio per factor level (Table 2, Figure 1) gives an unambiguous ranking: *population* ($\Delta = 0.207$ dB) and *iteration budget* ($\Delta = 0.166$ dB) dominate, while archive size (0.043) and β (0.067) lie within run-to-run noise; both influential effects *saturate* (most of the gain is captured by $N=50$ and $T=300$). The additive model predicts the optimum $N=70$, $T=500$ at $S/N \approx 125.19$ dB; a confirmation run gives 125.12 dB (HV = 1,803,110), validating the model. The 30-run study adopts $N=50$, $T=500$, archive 100, $\beta=1.5$, whose confirmation gives S/N 125.08 dB—within 0.47 % hypervolume of the L9 optimum, a deliberately conservative, compute-economical point on the S/N plateau ($T=500$ far exceeds the iteration-134 convergence point).

6 Results

6.1 Combined Pareto front and domination of FIFO

Combining and re-filtering the 30 runs yields a front of **406 non-dominated solutions**. Figure 2 shows it in the full objective space and Figure 3 in the f_1 – f_2 projection; in both, the FIFO baseline floats *above* the front, visual evidence that it is dominated. The extreme solutions (Table 3) expose the trade-offs.

★ FIFO baseline (dominated)

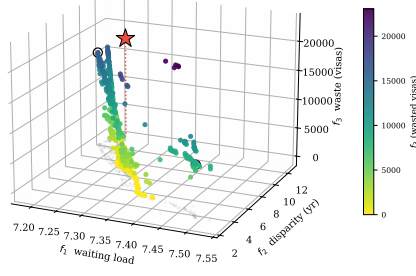


Fig. 2. The 406-solution combined Pareto front over the three objectives, colored by the waste f_3 . The FIFO baseline (red star) floats above the front—the dotted drop-line marks its distance to the front’s waste level—and is dominated; circles mark the extreme solutions.

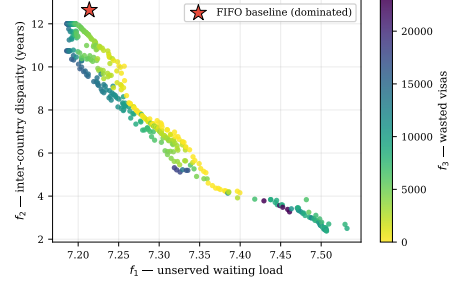


Fig. 3. f_1 – f_2 projection of the same front, colored by f_3 (wasted visas). FIFO lies inside the dominated region.

Table 3. Extreme solutions of the combined front, compared with the FIFO baseline. The minimum- f_1 solution strictly dominates FIFO on all three objectives.

Solution	f_1	f_2 (yr)	f_3 (visas)	Remark
Min. f_1 (least waiting)	7.1857	10.7436	16,280	Dominates FIFO
Min. f_2 (most equitable)	7.5067	2.3792	8,664	Radical equity
Min. f_3 (full use)	7.2572	9.3611	0	100 % utilization
FIFO baseline	7.2138	12.6377	17,540	Reference (dominated)

The central empirical finding is that **FIFO is not Pareto-optimal**: the minimum- f_1 solution improves f_1 from 7.2138 to 7.1857, f_2 from 12.64 to 10.74 years, and f_3 from 17,540 to 16,280 visas *simultaneously*. Moreover, 38 solutions of the front reach $f_3 = 0$ (zero waste), which FIFO never attains.

6.2 Convergence and robustness

The mean hypervolume reaches 95 % of its final value by iteration 9 and 99 % by iteration 134; the last 200 iterations add under 0.5 %. Over the 30 runs the HV distribution is narrow (mean 1,785,803, CV 1.56 %), evidencing a stable, reproducible algorithm.

6.3 Decoder, representation, and search: a controlled ladder

We attribute performance to its true source by climbing a ladder of methods, all sharing the decoder and a 25,000 function-evaluation budget—the comparison

currency (Table 4, Figure 4); the permutation-native methods inherit the Taguchi-tuned N and T and are not separately tuned, so any edge is not a tuning artifact. Under this budget MOHHO already beats the real-coded NSGA-II by 7.2 % in mean hypervolume (one-sided Mann–Whitney $p \approx 1.6 \times 10^{-10}$, $A_{12} = 0.97$), with $7.4\times$ smaller IGD and lower spacing.

The decoder does most of the work. Random restart—blind sampling with no search—already reaches a mean HV of 1,776,440, about 99.5 % of MOHHO’s and, strikingly, 6.6 % above NSGA-II. The feasibility-preserving decoder shapes a landscape on which even random sampling is strong; classic MOHHO improves on it only marginally (+0.5 %; paired Wilcoxon $p = 0.028$, marginal under Bonferroni), and giving NSGA-II the *identical* archive barely helps it (1,671,109 vs. 1,666,143), so MOHHO’s edge over NSGA-II is not an archiving artifact.

The representation–operator match is decisive. Why does a tuned NSGA-II fall below blind sampling? We measured the cause directly. On the SPV random keys, SBX and polynomial mutation are *near-identity on the decoded permutation*: over 3,000 trials the Kendall rank correlation between a parent’s SPV order and its child’s is $\tau = 0.99$. The GA perturbs the keys so little that the induced permutation barely changes, leaving it almost frozen in combinatorial space—precisely why it sinks below blind sampling, which at least draws fresh permutations. HHO’s position updates, by contrast, induce large permutation jumps ($\tau \approx 0$), so even on random keys the swarm genuinely traverses the decoded space (hence MOHHO > NSGA-II within the random-key tier). The clean fix, though, is to match the operators to the representation. Acting *directly* on permutations lifts *three different paradigms* above every random-key method: permutation-NSGA-II (dominance/GA) reaches 1,816,241, permutation-MOEA/D (decomposition) 1,814,179, and our **Discrete-MOHHO** (swarm) 1,808,850. The three cluster within 0.4 % of one another, yet all sit 1.3–1.7 % above the best random-key method (classic MOHHO)—several times the spread *among* the matched paradigms. Discrete-MOHHO improves on classic MOHHO by +1.3 % (paired Wilcoxon $p = 5 \times 10^{-7}$, better on 25/30 seeds) and is the *most stable* method (CV 0.46 %, versus 0.61 % for permutation-NSGA-II and 1.56 % for classic MOHHO), making it the most dependable choice for a single-run decision. The marginal mean lead belongs to permutation-NSGA-II (+0.4 % over Discrete-MOHHO, $A_{12} = 0.26$).

An empirical regularity. The ladder (Figure 4) makes the pattern unmistakable: performance is governed by the *feasibility-preserving decoder* and the *representation–operator match*, not by the metaheuristic family. That encoding–operator mismatch can hurt search is known in principle since random keys were introduced [2]; our contribution is the *controlled, quantified* demonstration on a real constrained problem—including the counterintuitive result that a tuned NSGA-II falls below blind sampling—and the decomposition of performance into decoder, representation, and family. That three *distinct* paradigms (swarm,

Table 4. The method ladder: 30 runs each, identical greedy decoder and 25,000-evaluation budget. Top block: random-key encoding. Bottom block: permutation-native, spanning three distinct paradigms (swarm, decomposition, dominance/GA), which cluster within 0.4 % and dominate every random-key method. Best per column in bold; “perm-” abbreviates “permutation-”.

Method	Paradigm	Mean HV $\pm \sigma$	CV	Comb. HV	# sols
NSGA-II	GA	1,666,143 $\pm$ 47,289	2.84 %	1,811,489	173
Random restart	—	1,776,440 $\pm$ 17,438	0.98 %	1,821,200	326
MOHHO	swarm	1,785,803 $\pm$ 27,831	1.56 %	1,831,654	406
Discrete-MOHHO	swarm	1,808,850 $\pm$ 8,264	0.46 %	1,839,510	396
perm-MOEA/D	decomp.	1,814,179 $\pm$ 8,396	0.46 %	1,841,569	366
perm-NSGA-II	GA	1,816,241 $\pm$ 11,040	0.61 %	1,852,415	372

decomposition, dominance/GA) converge to within 0.4 % once the representation is matched, all far above the random-key methods, is strong evidence that the family is second-order here. We are precise about what this shows: the three matched paradigms share the OX + swap operators by design, so the dimension we vary—and find second-order—is the *selection/replacement architecture* (Pareto sorting, scalarized decomposition, or energy-driven besiege) given a representation-matched operator set; the operator and the representation themselves are *first-order*, as the random-key tier attests. The matched swarm reclaims competitiveness and the best stability, which is why we promote Discrete-MOHHO. An omnibus Kruskal–Wallis test across the six methods rejects equal performance decisively ($H = 136.4$, $p \approx 10^{-27}$), and the mean ranks split into two clean tiers—the three permutation-native methods (best) above the three random-key methods. The ranking is also robust to the hypervolume reference point (five reference points; ordering preserved throughout). We frame this as an empirical regularity for this problem, not a general law: all instances here share one structural skeleton—which the second, structurally distinct problem of Section 6.5 directly addresses.

6.4 Generalization across instances

To check robustness to the demand estimate, we built five further instances perturbing every group’s demand by an independent uniform ± 20 % (fixed seeds) and recomputing the spillover-adjusted category and per-country caps. On all five—as on the base instance—FIFO is strictly dominated, MOHHO outperforms NSGA-II (one-sided Mann–Whitney $p \leq 1.3 \times 10^{-4}$ each), and the ordering MOHHO > random restart > NSGA-II holds (random restart 97–99.5 % of MOHHO’s HV, NSGA-II 91–93 %). This is robustness to demand perturbation; structural generalization is addressed next.

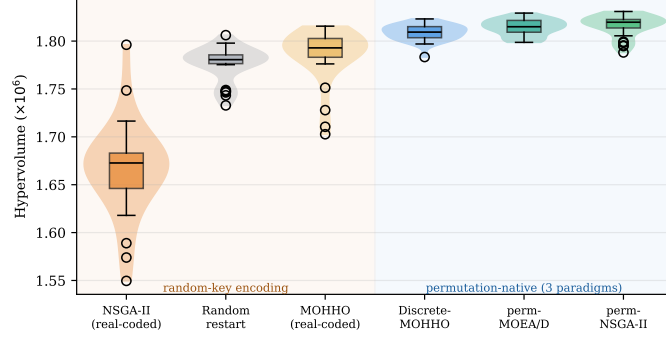


Fig. 4. The method ladder: per-run hypervolume (30 runs each, identical decoder and budget). Crossing from random-key encodings (left, shaded amber) to permutation-native operators (right, shaded blue) produces a clear jump; the metaheuristic family (GA vs. Harris Hawks) matters far less than the representation. Discrete-MOHHO is the tightest distribution (most stable).

6.5 Generalization to a structurally distinct problem

We replicate the ladder on a structurally distinct problem: a three-objective multidimensional 0/1 knapsack (MOMKP; 120 items, four capacity constraints, no caps, spillover, or waiting objective) under the same feasibility-preserving SPV + greedy decoder and the same six methods (30 seeds, paired Friedman test). The core finding replicates (Figure 5; Friedman $\chi^2 = 128.6$, $p = 4.7 \times 10^{-26}$, Nemenyi critical difference 1.38): the real-coded NSGA-II is again *worst* (mean rank 5.77, below random restart at 5.23) and permutation-NSGA-II again best (rank 1.20, +32 % over its real-coded form). The clean two-tier split does *not* fully transfer—a real-coded Harris Hawks optimizer is competitive here (rank 2.67). This is not an operator-order effect (τ is problem-independent: SBX 0.99, HHO ≈ 0 on both), and a capacity sweep *rejects* the natural “search headroom” explanation (Spearman $\rho = -0.90$). We therefore scope the regularity honestly: the representation–operator match dominates for interpolating, crossover-based search on both structures, whereas a method’s tolerance to a mismatched encoding is structure-specific and remains open.

7 Discussion

The trade-off is structural. Two countries hold over 60 % of demand with 10–13-year waits; the 7 % ceiling forbids fully serving them, so reducing disparity (f_2) leaves unserved load ($f_1 \uparrow$) and equitable spreading idles saturated categories ($f_3 \uparrow$). The three extreme solutions (Table 3) are mutually non-dominated—genuine three-way conflict. We keep f_1 as a distinct objective (its range is $\sim 28\times$ narrower than f_2 ’s because demand 868,098 far exceeds the 140,000 visas) because

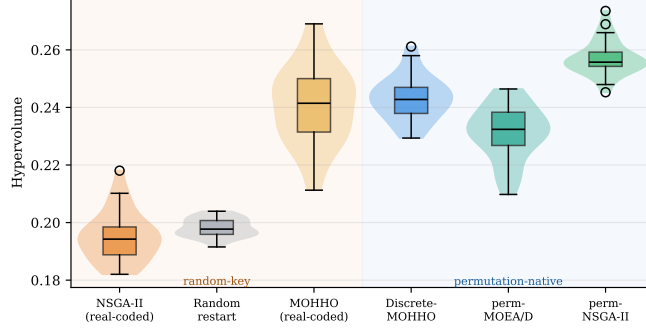


Fig. 5. The ladder on a structurally distinct problem (MOMKP). Real-coded NSGA-II and random restart stay at the bottom and permutation-NSGA-II at the top—as on the visa problem—but the real-coded Harris Hawks optimizer is competitive, so the clean two-tier split is structure-specific.

it is the minimum- f_1 policy that strictly dominates FIFO, which a two-objective reduction would obscure.

What governs performance, and policy. The ladder shows performance is governed by the feasibility-preserving decoder and the representation-operator match, not the metaheuristic family: a tuned real-coded GA falls below blind sampling (its operators are near-identity on the decoded permutation, $\tau = 0.99$), while permutation-native operators lift three paradigms to within 0.4 % of one another. The guidance is to invest in the decoder and representation before the search heuristic, and to prefer the most stable optimizer (Discrete-MOHHO) for single-run decisions. For policy, the front is a quantified menu: reducing disparity from 12.64 to 2.38 years costs only $\sim 4\%$ in f_1 relative to FIFO [7,10]. Each front solution maps to a concrete allocation in which India and China stay at their 7 % ceiling while visas FIFO leaves in the residual pool are redirected to underserved nationalities.

Limitations. The model covers a single fiscal year, uses 21 representative countries, estimates demand from public aggregates [18,19], and assumes every assigned visa is used. These affect external validity, not the internal comparison among the six methods, which share data, decoder, and budget.

8 Conclusions

We formulated the annual allocation of U.S. employment-based visas as a three-objective integer program with six hard constraints and solved it through a feasibility-preserving SPV + greedy decoder that guarantees feasibility by construction—no penalties, no repair. A Taguchi L9 design replaced empirical parameter choices with a confirmed operating point. Across 30 independent

runs the recovered Pareto front strictly dominates the FIFO incumbent on all three objectives, with 38 zero-waste solutions. Our central result comes from a controlled six-method ladder: the decoder alone (random restart) already beats a real-coded NSGA-II, and—the lesson—the representation-operator match, not the metaheuristic family, governs performance. Matching operators to the representation lifts three distinct paradigms (a GA, a decomposition method, and a Harris Hawks optimizer) to within 0.4% of one another and above every random-key method; the Discrete-MOHHO we introduce improves on classic MOHHO by 1.3% ($p = 5 \times 10^{-7}$) and is the most stable optimizer of all (CV 0.46%), within 0.5% of the best. All conclusions hold across five perturbed-demand instances, and on a structurally distinct multidimensional knapsack the core finding—a tuned real-coded GA falling below random sampling, reversed by representation-matched operators—replicates, while the two-tier split is structure-specific: operator order-preservation (problem-independent, $\tau=0.99$) explains the robust GA-freeze, but the cause of the random-key swarm’s structure-specific competitiveness resists a simple search-headroom explanation—which we test with a capacity sweep and reject—and we report it as an open question. Beyond the numbers, the method yields an auditable menu of allocation policies rather than a single recommendation. Future work includes a multi-period formulation, the full country roster, further problem domains beyond the two studied here, and additional optimizers such as SPEA2.

Data and Code Availability. The problem data, the optimizer, the Taguchi design, and the scripts that reproduce every figure and table are available at https://github.com/jrebull/MIAAD_Harrisv2.

Acknowledgments. The authors thank the Maestría en Inteligencia Artificial y Analítica de Datos at the Universidad Autónoma de Ciudad Juárez for institutional support.

References

1. Al Rafei, T., Yousfi Steiner, N., Chrenko, D.: Genetic algorithm and Taguchi method: an approach for better Li-ion cell model parameter identification. *Batteries* **9**(2), 72 (2023). <https://doi.org/10.3390/batteries9020072>
2. Bean, J.C.: Genetic algorithms and random keys for sequencing and optimization. *ORSA Journal on Computing* **6**(2), 154–160 (1994). <https://doi.org/10.1287/ijoc.6.2.154>
3. Çerçi, S., Dönmez, E.: Hybrid multi-objective Harris Hawk optimization algorithm based on elite non-dominated sorting and grid index mechanism. *Advances in Engineering Software* **173**, 103214 (2022). <https://doi.org/10.1016/j.advengsoft.2022.103214>
4. Chaves, A.A., Resende, M.G.C., Schuetz, M.J.A., Brubaker, J.K., Katzgraber, H.G., de Arruda, E.F., et al.: A random-key optimizer for combinatorial optimization. *Journal of Heuristics* **31**(4), 32 (2025). <https://doi.org/10.1007/s10732-025-09568-z>

5. Choo, Y.H., Cai, Z., Le, V., Johnstone, M., Creighton, D., Lim, C.P.: Enhancing the Harris' Hawk optimiser for single- and multi-objective optimisation. *Soft Computing* **27**(22), 16675–16715 (2023). <https://doi.org/10.1007/s00500-023-08952-w>
6. Coello Coello, C.A., Lamont, G.B., Van Veldhuizen, D.A.: *Evolutionary Algorithms for Solving Multi-Objective Problems*, 2nd edn. Springer, New York (2007). <https://doi.org/10.1007/978-0-387-36797-2>
7. Congressional Research Service: U.S. employment-based immigration policy. Report R47164 (2023), <https://www.congress.gov/crs-product/R47164>
8. Deb, K., Pratap, A., Agarwal, S., Meyarivan, T.: A fast and elitist multiobjective genetic algorithm: NSGA-II. *IEEE Transactions on Evolutionary Computation* **6**(2), 182–197 (2002). <https://doi.org/10.1109/4235.996017>
9. Derrac, J., García, S., Molina, D., Herrera, F.: A practical tutorial on the use of nonparametric statistical tests as a methodology for comparing evolutionary and swarm intelligence algorithms. *Swarm and Evolutionary Computation* **1**(1), 3–18 (2011). <https://doi.org/10.1016/j.swevo.2011.02.002>
10. FWD.us: Per-country cap reform: priority bill spotlight (2024), <https://www.fwd.us/news/per-country-cap-reform-priority-bill-spotlight/>
11. Garey, M.R., Johnson, D.S.: *Computers and Intractability: A Guide to the Theory of NP-Completeness*. W.H. Freeman, San Francisco (1979)
12. Heidari, A.A., Mirjalili, S., Faris, H., Aljarah, I., Mafarja, M., Chen, H.: Harris hawks optimization: algorithm and applications. *Future Generation Computer Systems* **97**, 849–872 (2019). <https://doi.org/10.1016/j.future.2019.02.028>
13. Jangir, P., Heidari, A.A., Chen, H.: Elitist non-dominated sorting Harris hawks optimization: framework and developments for multi-objective problems. *Expert Systems with Applications* **186**, 115747 (2021). <https://doi.org/10.1016/j.eswa.2021.115747>
14. Kuşoğlu, İ., Yüzgeç, U.: Multi-objective Harris Hawks optimizer for multiobjective optimization problems. *BSEU Journal of Engineering Research and Technology* **1**(1), 1–12 (2020)
15. Londe, M.A., Pessoa, L.S., Andrade, C.E., Resende, M.G.C.: Biased random-key genetic algorithms: a review. *European Journal of Operational Research* **321**(1), 1–22 (2025). <https://doi.org/10.1016/j.ejor.2024.03.030>
16. Montgomery, D.C.: *Design and Analysis of Experiments*, 9th edn. Wiley, Hoboken (2017)
17. Rahimi, I., Gandomi, A.H., Chen, F., Mezura-Montes, E.: A review on constraint handling techniques for population-based algorithms: from single-objective to multi-objective optimization. *Archives of Computational Methods in Engineering* **30**(3), 2181–2209 (2023). <https://doi.org/10.1007/s11831-022-09859-9>
18. U.S. Citizenship and Immigration Services: Fiscal year 2023 employment-based adjustment of status FAQs (2023), <https://www.uscis.gov/>
19. U.S. Department of State: Visa bulletin for February 2026 (2026), <https://travel.state.gov/content/travel/en/legal/visa-law0/visa-bulletin.html>
20. Vargha, A., Delaney, H.D.: A critique and improvement of the CL common language effect size statistics of McGraw and Wong. *Journal of Educational and Behavioral Statistics* **25**(2), 101–132 (2000). <https://doi.org/10.3102/10769986025002101>
21. Zhang, Q., Li, H.: MOEA/D: a multiobjective evolutionary algorithm based on decomposition. *IEEE Transactions on Evolutionary Computation* **11**(6), 712–731 (2007). <https://doi.org/10.1109/TEVC.2007.892759>
22. Zhang, W., Xiao, G., Gen, M., Geng, H., Wang, X., Deng, M., et al.: Enhancing multi-objective evolutionary algorithms with machine learning for scheduling problems:

- recent advances and survey. *Frontiers in Industrial Engineering* **2**, 1337174 (2024). <https://doi.org/10.3389/fieng.2024.1337174>
23. Zitzler, E., Thiele, L.: Multiobjective evolutionary algorithms: a comparative case study and the strength Pareto approach. *IEEE Transactions on Evolutionary Computation* **3**(4), 257–271 (1999). <https://doi.org/10.1109/4235.797969>